A CRITERION FOR INJECTIVITY OF A MODULE MAP IN TERMS OF ASSOCIATED PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Description of an associated prime of a module through an element of the localisation whose annihilator is the prime. Algebra, Lemma \ref{algebra-lemma-associated-primes-localize}, cited at proof lines 5443-5445. Produces the element contradicting the hypothesis.
- A module with no associated primes is zero. Algebra, Lemma \ref{algebra-lemma-ass-zero}, cited at proof lines 5448-5449. Concludes that the kernel vanishes.

Source reference. Original lemma 0AV7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13675–13675. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0310: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3967–3967. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0587: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6770–6770. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Let $\varphi : M \to N$ be a map of R-modules. Assume that for every prime $\mathfrak{p}$ of R at least one of the following happens*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $M_\mathfrak{p} \to N_\mathfrak{p}$ *is injective, or*
(2) $\mathfrak{p} \notin \mathit{Ass}(M)$.

Then φ is injective.

Proof. Let $\mathfrak{p}$ be an associated prime of $\mathrm{Ker}(\varphi)$. Then there exists an element $x \in M_\mathfrak{p}$ which is in the kernel of $M_\mathfrak{p} \to N_\mathfrak{p}$ and whose annihilator is $\mathfrak{p}R_\mathfrak{p}$ (Algebra, Lemma 0310). This is impossible in both cases. Hence $\mathrm{Ass}(\mathrm{Ker}(\varphi)) = \emptyset$ and we conclude $\mathrm{Ker}(\varphi) = 0$ by Algebra, Lemma 0587. □